KOI FALSE-POSITIVE PROJECT · MIND MAP · STUDENT WALL MAP

Can a computer spot a fake planet?

Everything the notebook asks you to do, on one page.

STAGE A
WEEK 1 · Notebook Part 1

The catalogue, and the clue in it

- Download the KOI table from the NASA Exoplanet Archive. Keep CONFIRMED and FALSE POSITIVE, drop CANDIDATE.
- Plot depth, period and impact parameter before building anything.
- Work out the guessing baseline: about 47% are false positives, so accuracy on its own means very little.
- **Run the leakage check and set aside nine columns that were filled in after somebody knew the answer.**
- Train a Random Forest on the eight columns that are left, splitting by star.
- **Put koi_score back on purpose. Accuracy jumps to 1.00.**
- **A very high score is a reason to look for a mistake.**

THE SIX PARTS YOU WRITE YOURSELF

TODO 1	`odd_even_sigma`	combine boolean masks
TODO 2	`secondary_ppm`	phase runs -P/2 to +P/2, so half a period away is at both ends
TODO 3	`rho_star_cgs`	unit handling; work in seconds
TODO 4	`grouped split`	did the leakage lesson transfer?
TODO 5	`scatter plot`	read a two-class plot honestly
TODO 6	`flag comparison`	turn a confusion matrix into physics

STAGE B
WEEK 1 · Notebook Part 2

From brightness to numbers

- Take 50 confirmed planets and 50 false positives with a fixed random seed, and download with author="Kepler" so all come from one pipeline.
- **Clip spikes upward only. The symmetric setting would delete deep eclipses.**
- **Detrend with flatten(mask=...) so the fit ignores the in-transit points.**
- Measure in three windows: core 0.30, full 0.50, baseline 1 to 4 x duration.
- **Test the functions on a transit you injected yourself before trusting real data.**
- Run the loop. It saves progress every ten targets, so it can be restarted.

THE TEN FEATURES YOU MEASURE

★ = you write it

- `depth_ppm`
- `n_transits`
- `snr`
- ★ **`odd_even_sigma`**
- `depth_scatter`
- `duration_hr`
- `scatter_oot_ppm`
- `v_shape`
- ★ **`secondary_ppm`**
- ★ **`rho_star_cgs`**

THE BIG IDEA

A light curve records how much light was lost, and when.

It never records where on the sky the light was lost.

So a pair of eclipsing stars near your target makes an identical dip.

That kind of false positive can never be found from brightness alone.

Measuring where that limit sits IS the scientific result.

THE QUESTION

Can a computer spot a fake planet?

What fraction of Kepler's false positives can be identified from the shape of the light curve alone?

FOUR MISTAKES TO AVOID

each one makes the result look better than it is

1. **Using a column that holds the answer**
   koi_score, the four fpflag columns and kepler_name were all written after the verdict was decided.
2. **Splitting by row instead of by star**
   Some stars have several KOIs. Group the split on kepid or the score comes out too high.
3. **Judging the model by its accuracy**
   Half the objects are fakes, so always answering 'fake' scores about 50%. Use precision and recall.
4. **Removing the signal with the noise**
   Too strong a filter, or a symmetric outlier cut, eats a deep eclipse. Mask the transit, then inject a fake one to check.

STAGE C
WEEK 2 · Notebook Part 3

Training the model

- From here on every number came from the brightness data. No catalogue columns are used.
- **Split by kepid, never by row, so one star cannot sit on both sides.**
- **Read the recall for the false-positive class first. That is the number this project is about.**
- Draw the precision-recall curve with the guessing line marked, not a ROC curve.
- Repeat with grouped 5-fold cross-validation for a fairer score.
- Rank the feature importances, then plot the two strongest against each other.

HOW YOU SCORE THE MODEL

accuracy	Not useful here. Half the objects are fakes, so answering 'fake' every time already scores about 50%.
precision	Of the objects you called fake, what fraction really were fake.
recall	Of all the fakes, what fraction you caught. This is the number the project is about.
PR curve	Precision against recall at every threshold, with the guessing line drawn on for comparison.
grouped 5-fold CV	Repeats the whole train-and-test cycle five times so every object is tested exactly once.

STAGE D
WEEK 2 · Notebook Part 4

Finding the limit

- Attach the four koi_fpflag_* columns. Analysis only, never training.
- Split the true false positives into the ones caught and the ones missed.
- Work out what fraction of each group carries each flag.
- **Check whether the centroid-offset flag dominates the objects you missed.**
- **Centroid offset is measured from the camera pixels, not the light curve, so no feature of yours could have caught it.**
- Write the answer as a number with an uncertainty, compared with guessing, and with a physical reason.

THE ANSWER YOU ARE AIMING FOR

Using ten features measured from Kepler light curves, a Random Forest identified XX% +/- Y% of false positives at a precision of ZZ% (grouped 5-fold cross-validation, N = ___). Of the false positives the model missed, AA% carried the centroid-offset flag, against BB% of those it caught. A centroid offset is measured from the camera pixels, not the light curve, so this type cannot be identified from brightness alone.